

Video Transcripts

Module 18: Platforms for Handling Big Data

Video 1: Introduction (00:39)

Working with big data is challenging. You have volume, you have variety, and you have velocity. The platforms of the past really weren't equipped to do that. Now there's a new generation of big data and distributed computing platforms that allow us to do that. They have looked at data storage and processing. So, let's go ahead and jump in and look at the first one that really changed the paradigm when it came to scalability, flexibility, and broke the pattern of the old applications and provided us with this new resource.

Video 2: Overview of Hadoop: A Powerful and Extensible Platform (08:16)

Hadoop is one of those projects that changes the course of computing, especially when it comes to the data side of computing. The early days previous to the web were days in where, when it came to data, there was a lot less complexity. A lot of the organizations in the enterprises had data that was pretty well-structured in its generation and so storage fit neatly into rows and columns mostly as you can see. And so you had a world where you had a lot less velocity, variety, and volume. Now after the world goes through the revolution of the web, you certainly have a tremendous amount of variety.

The two way channel that is the web, allowed anyone to be able to produce data and certainly to intake data as well. So, that is an explosion. You have much more variety, you have velocity, the amount of data that is being produced is certainly overwhelming and you have a tremendous amount of volume as well. So, all of that doesn't fit neatly anymore into that rows and column, neat picture of the past and we really entered the era of big data that has very different needs. Now a company that was looking at this at the time was Google.

They were faced with the challenge of being able to parse the information that existed in the web, certainly a great deal of variety of velocity and volume. And so, the machines and the systems and the platforms that existed to be able to address this need simply didn't stretch to that. And so they needed to come up with a different way of doing this and they wanted to

leverage commodity machines that they could simply throw all of this data into and from that a programming model that allowed you to be able to process those great deal big volumes of data.

And so they introduced a number of innovations into the market. And especially in this case, I want to emphasize Google File System, the fabric that stores all of that information, MapReduce, the programming model that allows you to process all of that data. And they added additional layers that are not as relevant to what we will be discussing here today when it comes to Hadoop. So, Hadoop tries to take those ideas. The software that Google had built was proprietary, but they shared their innovations and their ideas in carefully detailed publications.

One for the Google File system and other one for MapReduce, they did another one for Big Table, another layer. And, these were then used and they inspired a new generation of software engineers and data scientists and data engineers to build this new platform based on those ideas. And that idea is called Hadoop. And as you can see here, the very first layer of it was the distributed file system, which is the equivalent to Google's file system, GFS in this case called HDFS, to be able to store those volumes of data. And as you can see here, the fabric is responsible for taking those inputs of data, breaking them down into blocks and mapping them onto machines.

As you can see here, there is that varied or rich or unstructured data that then gets mapped onto those machines. And you can see there the block size that the default there being 128 megabytes and how that would be broken down. Now, it's worth noting that this was also or part of this platform was also the redundancy that was needed to be able to afford those failures in that commodity hardware. So, you can see there, the computing fabric was handling this. And in fact, one of the number that became well known was that there would be three copies made in case one of those failed.

And it turned out to be a good one, and one that has held for a long time now. The next layer was called the exact same thing as Google's, which was MapReduce, and that is, that model to be able to carry out computation. And this is one that existed even, or the idea of it existed even prior to Google, that you have a list of data onto which you're going to map functions, and then the output of those functions of each of the items of that data are going to be reduced onto a result.

And so here is a practical example of how you might do that. And in this case, this one's specific to Hadoop. And you can see there that input file from the famous book, it was the best of times,

it was the worst of times, and so on. And how that could be a split there onto that list. Then, you map the functions onto it. And in this case, we're counting the frequency of words. And you can see there the values of the output of the mapping of the functions. Then, we would do a sort, and finally a reduction where we simply add up all the counts of those words, and you can see there the final result.

Now, this turns out to be the foundation upon which a great deal more is built. But even at this level, you can see here that you have a tremendously scalable resource that the world really didn't have and it was free, it was open-source, and it was shared openly. The last layer there is one that I didn't mention, the YARN, yet another resource negotiator, and you can think of that as the management for this platform. So, what is it? More concisely, it's a scalable platform for distributed storage and distributed processing. And the next part of the definition there is the important one, that it is made for very large datasets. Really, this has no limitations.

You can simply throw hardware at it as your needs increase. And this scales horizontally, meaning that you simply need to add more machines which is a great thing. Now the other thing to note as well is that this was a big shift in how things had been done up until that point. The platforms that organizations had were very focused at a few specific use cases. The data warehouses at the time certainly were that. And here, you have a very different model, one that allows you to act upon tremendously large datasets. And as you can see there in the second block, increased computing power, fault tolerance, flexibility in data management.

That is, you could come up with any type of application that you wanted beyond the cookie-cutter solutions that existed in the data warehousing days at those times. And you had now a tremendous amount of resources that you could pick off the shelf that didn't require you to buy a license, that were open-source, and we see that shift that takes place in the industry. Nowadays, that is commonplace, that the biggest companies in the world, companies like Facebook, Amazon, many others are powered by open-source. But in those days, you started to see one of the first chefs.

You have this tremendous new platform that is robust, that is reliable, that is being used heavily in industry. And so, as you can see here, that becomes part of the story when it comes to big data needs in a platform to address them. You can still see here that even though this has been extended to capture a lot of the packages that are now new and that have emerged since then, you still have that footprint, you have HDFS, you have MapReduce, you have YARN, and then you

have a host of other ones that are using those building blocks. But through it, you now have data storage, data processing, access and management. All of it in that new open-source paradigm. All of it addressing the multitude of needs when it comes to big data and big data needs.

Video 3: Creating Hadoop Docker Images (01:52)

As we have done in the past, we're going to be using Docker to simplify our process in creating an instance of Hadoop. And in this case, we're going to be using images, that is Docker images that have been created by Big Data Europe's effort. So, let's go ahead and navigate to the site. I'm first going to navigate to the project address, as you can see, and then afterwards, I will be looking at the repository. So, let's go ahead and paste that address into the browser. And as you can see there, that's Big Data Europe integrating big data software in communities for addressing their societal challenges.

Now, it's worth noting that this big platforms have led the way to promote collaboration, to be able to Dockerize, containerize this build building blocks, to be able to use these platforms, in this case GitHub, to be able to bring together those efforts. That's a great thing for everybody involved. And obviously the exact same approach can be taken in business organizations as well. So, let's go ahead and take a look at the project. Let's go ahead and paste the repo address.

And as you can see here, you have docker-hadoop. This is a project that is maintained by them and that once you download, you will be able to run the Docker compose file, as you see here. So, now I'll leave you to do that. Go ahead and download that project. You can either download here through the zip or you can simply clone the project onto your desktop, which is what I did. It's a simpler way to do it if you have git installed on your machine.

Video 4: Creating Hadoop Docker Containers (02:14)

It is now time to create our containers. Once you download the project, go ahead and move into that directory and run docker-compose up. I'm going to be doing the same now myself. And as you can see here, I am within that directory. If I list my files, you can see there my docker-compose. So, now I'm going to be running that command. But before I do that, let me go ahead and show you my dashboard. And as you can see here, I have no containers running. However, I have run it previously and the images have been downloaded to my machine and I will be taking advantage of that.

So, when I do docker-compose up, my creation of the Hadoop environment will be faster because the images have been cached. So, I'll go ahead and enter docker-compose up. So, as you can see there, I have all of the pieces up and running and if we take a look at our dashboard, I now have an entry and if I expand that, you'll see there that I have the different pieces. So, I have a nodemanager, a resource manager, I have historyserver, a namenode, and a datanode. And if I list my files or my containers, you can see there that we'll see the same. And the names are resource manager, namenode, nodemanager, historyserver, and datanode.

And so, to emphasize what we have there, the very first one that I have there on the list is HDFS, this is the file system, the primary node to manage the secondary. Then, we have 3 datanodes, these are the secondary nodes I just mentioned. Then we have YARN and a history and a manager. So, make sure you can go through this setup and that you have all of these containers up and running before you go forward.

Video 5: Checking Hadoop Status Using a Web Browser (00:46)

Before we go forward, let's go ahead and take a look at the current status. We can view it by navigating to the following address and port. So, as you can see here I have the summary. You can see the number of nodes that you have running. If you navigate through the menus in the top bar, you can take a look at a little bit more detail. For example, here is one for the data nodes. And you can see here that we haven't yet started using it, so there's really not that much to see. But this is a good place to come back and look for clues if something's not working like you think it should. So, once you have confirmed this, let's go ahead and move on to the next section.

Video 6: Creating Input for a Hadoop Container (02:13)

To get some experience with Hadoop, we're going to be running a word frequency, a word counting program. We're going to start off by logging into the container, generating some input data, later downloading a program, and after that we're going to be running all of that. So, we'll start off here by accessing the container and creating some of the input. There are a few ways in which you can access that container, but in this case, we're going to take the easy way. We're going to use the graphical user interface, and I'm going to go ahead and select here 'CLI' for command line interface, and I'll go ahead and click it. And as you'll see, it'll open up a new window.

Now that we're inside of it, let's go ahead and list our files. That's just simply a good practice. Then we're going to go ahead and make a directory, and the directory will be input, a directory for input. We're then going to echo into a file inside of input called f1. It was the best of times. It was the worst of times. Go ahead and do that. And now we have that file inside of input. I'm now going to create a second file as you can see there, f2. And if we were to move inside of input, you'd see that we have those two files as expected.

Next, we want to be able to move that content into HDFS, that is the Hadoop file system. I'm going to start off by creating a directory, as you can see there, called input inside of Hadoop. Next, to move our input into the data nodes on HDFS, we're going to run the following, which is basically going to copy our input into HDFS. And so at this point, to review, we have created some input files. We've then taken those input files, and pushed them into the file system, Hadoop's file system. So, we're now ready to take another step.

Video 7: Setting Up a Hadoop Container to Run a Word Count Program (02:08)

To count the word frequency on our input, we're going to need a program. Now, we could write that program or we could download it. And this is one of those things that is often done when you stand up one of these platforms and so as you can imagine, there are many of them that have been written. Now, you could download it onto your local machine, your host, and then upload it onto your container. However, in our case, we're going to download it directly from within the container.

We're going to be running curl. As you can see there, the -L option is one you might have not seen before, and this one simply follows the link in case there are any redirects from that initial file that you point to. Then as you can see there, we have an option as well to output and some file name that we're going to give it. So, the address that we're going to be targeting there is one of a repository for Java type programs, you can see there, the long link in blue in that the name of that file is 'hadoop-mapreduce-examples,' a number, and then .jar. And we're going to be saving it as that name as well.

So, let's go ahead and move into the container and download it. So, here we are inside of the container, and I'm going to go ahead and paste that instruction and you can see there that it's the same one that I had on the slide. So, I'm going to go ahead and hit return and you can see there some of the feedback of the data. Now, if I go ahead and list the files, you'll see there that

we have there the long file name and that file is now in our in our file system. So, if I list it, you can see there that's perhaps easier to see in that listing. So, go ahead and do the same. Download that file into your container as mentioned before. If you prefer, you can download it to your file system and then from there push it into the container.

Video 8: Executing the Word Count Program and Inspecting the Results (01:35)

It is now time to run our word frequency program and we're going to be running that program on the Hadoop platform. You can see there the instruction and the different components of that command or terminal instruction. So, let's go ahead and take that and execute it inside of our namenode container. I'm going to go ahead and clear the screen. I'm going to go ahead and put in the instruction. And as you can see there, that is going to run on Hadoop and we are specifying there the input folder and also the output folder.

So, I'm going to go ahead and hit return. And as you can see there that has run to completion. We could scroll back up to see all of the output that this instruction has generated. Now, to look at the results of the program, we can look at the output folder. I'm going to go ahead and clear the screen once again. And you can see there that we're reaching into HDFS. And as expected, there we have the word counts we were expecting. So, make sure you can run this last part, make sure that the numbers check out and that you can leverage the platform.

Video 9: Recap: Using Hadoop to Handle Big Data (02:17)

Hadoop is a powerful and extensible platform. More than that, it is also an open-source project that anybody can download and customize to any need that they have without needing to pay any licenses. It also marked a transition between the very cookie-cutter, very close form solutions that came before the days of the web where you had data that didn't vary that much to what happened afterwards. The tremendous flood of information that had high velocity, high variety, and high volume, and because of it, it really became a building block for a lot of the ecosystem of data products that followed that first three tiers that copied what Google initially did.

The inspiration of GFS to Google File System, Google's MapReduce and other tiers really led the way for what the authors of Hadoop implemented the HDFS, MapReduce and YARN as well. And now, as mentioned previously, it is part of a broader ecosystem of open source tools that are needing or are meeting the biggest needs that industry and government and educational

organizations throughout the world have. And so, it is a great platform to get to know, there's a tremendous amount to learn. We overviewed the biggest patterns, but we will leave it up to you to dig deeper if you would like to.

One fun exercise that you may want to do is to download an open-source book, you can see here that Project Gutenberg has the public domain books within the US. Those are available in text. So, you could download them, split them up into files, and run your word counting type of program. Now there's other many other things that you could do as well, but this is a fun one you want to try. So, with that in mind, let's go ahead and move forward, taking into account the great contributions that Hadoop has made to the big data problem.

